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1. Introduction

1.1. Scope of the paper

The aim of this article is to introduce, discuss and review the main aspects of femtosecond laser *filamentation* in various transparent nonlinear and dispersive media. In this review, we do not consider *relativistic filamentation*, corresponding to the propagation of a very intense ($I > 10^{17} \text{ W/cm}^2$) laser pulse in a fully ionized medium where electrons oscillate at a velocity approaching c , the light velocity in vacuum. We are concerned with filamentation in a continuous dielectric medium, which becomes weakly ionized by the propagating pulse, of typical peak intensity $\sim 10^{13} \text{ W/cm}^2$. The main emphasis will be given to air. However, we will also discuss filamentation in other gases and in condensed media such as transparent solids and liquids.

It has been long believed that intense ultra-short laser pulses are not suited for long range propagation in air. For example, in a linear propagation regime the peak intensity of a femtosecond pulse of initial duration $t_p = 30 \text{ fs}$ ($1 \text{ fs} = 10^{-15} \text{ s}$) with a beam waist of 5 mm is expected to be reduced by a factor of $\sim 5 \times 10^3$ after 1 km of propagation in air due to the combined effect of beam diffraction (factor of 100) and group velocity dispersion (factor of 50). However, experiments using an intense infrared (IR) femtosecond pulse show the opposite trend. In the first experiments carried out by Braun et al. (1995), the intensity of the pulse increased instead of decreasing. It reached sufficiently high values to provoke micro-burns on a hard coated mirror 10 m away from the laser. The same mirror was undamaged if placed immediately close to the laser output, before propagation in air. The researchers from Michigan University correctly explained that a nonlinear effect was responsible for the damage tracks on the mirror: This effect is beam self-focusing. The damage tracks subsisted if the mirror was displaced several meters along the propagation axis. The beam therefore remained focused over an extended distance. This type of propagation was called filamentation or self-guided propagation. Shortly afterwards, filamentation of femtosecond laser pulses was shown to occur over more than 50 m in Laboratoire d'Optique Appliquée, (see Fig. 1, Nibbering et al., 1996) and then over several hundreds of meters (La Fontaine et al., 1999b). Experiments in Spring 2003 at École Polytechnique have revealed an horizontal filamentation over a distance larger than 2 km (Méchain et al., 2004a, 2005a). Experiments of vertical propagation suggest even larger filamentation distances, (see Fig. 2, Wöste et al., 1997 and Rodriguez et al., 2004).

Filamentation aroused a great interest and became a field of intense research activity as shown from the reference list. It is worth stressing at this point that the term “filamentation” may be misleading in the context of ultrashort lasers because it suggests an extended light string whereas the pulse actually shortens in the propagation direction while the beam shrinks and maintains a narrow diameter in the transverse diffraction plane. The term of light bullets would be more appropriate (Silberberg, 1990), although filaments are neither frozen, nor localized structures in time or in space. Throughout this paper, we will nevertheless conform to the custom and continue to use the terms of *filaments* and *filamentation* to denote a dynamic structure with an intense core, that is able to propagate over extended distances much larger than the typical diffraction length while keeping a narrow beam size without the help of any external guiding mechanism. The definition used in the literature is sometimes more strict. The intensity of the core reaching 10^{13} – 10^{14} W/cm^2 is sufficient to ionize the molecules or atoms of the medium. The term *filament* is then used to describe the part of the propagation during which the pulse generates a column of weakly ionized plasma in its wake. In our general definition of a filament, ionization is not a necessary condition. Femtosecond filamentation was observed for various laser wavelengths from the ultraviolet to the infrared domain, and for various pulse durations from several tenth of fs to several ps. This generic phenomenon was also observed in transparent solids and liquids and does not critically depend on the initial conditions of the laser or the propagation medium.

1.2. Properties of ultrashort pulse nonlinear propagation

The physical origin of the formation of filaments in air is now well understood. Even if many physical effects come into play during the propagation of the pulse in the filament, the process of its formation can be described by the action

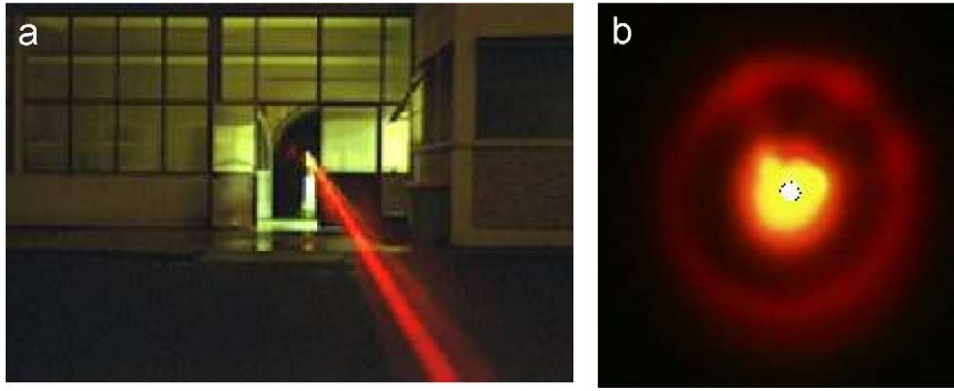


Fig. 1. (a) Propagation of an infrared laser beam (800 nm) in the Laboratory of Applied Optics (École Polytechnique, Palaiseau). The main properties of the filament are a hot core of diameter $\sim 100 \mu\text{m}$ and intensity $\sim 5 \times 10^{13} \text{ W/cm}^2$ generating an electron plasma of density 10^{16} cm^{-3} and a white continuum. (b) Image of the beam after a propagation over 50 m showing a white core in the center of the filament with a diameter of $100 \mu\text{m}$.



Fig. 2. Vertical propagation in the sky of the teramobile laser beam (peak power of 5 TW at 800 nm, see Section 5.1.1): backward scattering of the beam is recorded using the telescope with a 2 m aperture, located in Tautenburg, Germany. One distinguishes a zone of Mie scattering from a vapor cloud located at an altitude of 9 km.

of mainly two nonlinear physical effects: the optical Kerr effect on the one hand, acts against diffraction and tends to focus the beam on itself. On the other hand, multiphoton absorption limits the intensity. The ensuing ionization of the atmosphere reduces the local refraction index of the medium and leads to beam defocusing. In the following, we briefly review qualitatively these effects as well as other physical effects that play a role in femtosecond filamentation, either in gases or in condensed media. We start by discussing the linear effects, then the nonlinear effects entering the polarization of the medium, and finally the nonlinear effects linked to plasma generation.

1.2.1. Diffraction

Diffraction of laser beams always occurs, even if they propagate in vacuum. The aim of this paragraph is simply to introduce the typical length for diffraction of a laser beam. The laws of Gaussian optics say that the width of a Gaussian

beam with a flat spatial phase increases by a factor of $\sqrt{2}$ when the beam propagates over a typical length, called the Rayleigh length. This length is defined as

$$L_{\text{DF}} = \frac{k w_0^2}{2} = \frac{\pi n_0 w_0^2}{\lambda_0}, \quad (1)$$

where w_0 is the beam waist, λ_0 the laser wavelength in vacuum, n_0 the refraction index of the medium at this wavelength, $k \equiv n_0 k_0$ and $k_0 \equiv 2\pi/\lambda_0$, the wavenumbers in the medium and in vacuum, respectively. For example, the Rayleigh length of a beam with $w_0 = 100 \mu\text{m}$ at $\lambda_0 = 800 \text{ nm}$ in vacuum ($n_0 = 1$) is $L_{\text{DF}} = 3.9 \text{ cm}$.

1.2.2. Space–time defocusing

It is generally assumed that a laser pulse propagating in vacuum keeps its duration constant. However, this is not strictly the case. A laser pulse is not monochromatic. In a beam of finite dimension, bluer frequencies diffract less than redder frequencies. Therefore a detector (of limited size) will record a longer pulse after propagation because the spectrum of the pulse reaching the detector has become narrower.

1.2.3. Group velocity dispersion and higher order dispersive effects

Gases, liquids and transparent solids are dispersive media. In a region of normal dispersion, red frequencies are faster than blue frequencies. This means that after propagation, the redder frequencies of a pulse with an initial flat temporal phase will pile up in the leading part of the pulse envelope while the bluer frequencies will be found in the trailing part. This increases the pulse duration and decreases the peak intensity. This effect, called group velocity dispersion, is characterized by the dispersive length

$$L_{\text{GVD}} = \frac{t_p^2}{2k''}, \quad (2)$$

where t_p is the pulse duration and $k'' \equiv \partial^2 k / \partial \omega^2|_{\omega_0}$ denotes the coefficient of the quadratic term in the development of the wavenumber as a function of the departure from the central frequency ω_0 of the pulse: $k(\omega) = n(\omega)\omega/c = n_0\omega_0/c + k'(\omega - \omega_0) + k''(\omega - \omega_0)^2/2 + \dots$. For instance, a $t_p = 10 \text{ fs}$ pulse at 800 nm will increase its duration by 40% after only $\sim 2.5 \text{ m}$ in air ($k'' = 0.2 \text{ fs}^2/\text{cm}$) or 1.4 mm in glass ($k'' = 360 \text{ fs}^2/\text{cm}$).

1.2.4. Self-focusing

The refractive index of air n in the presence of an intense electromagnetic field does not only depend on its frequency, but also on the space and time dependent intensity $I(r, t)$ of the laser according to the law: $n = n_0 + n_2 I(r, t)$. The coefficient of the nonlinear Kerr index n_2 is related to the third order susceptibility $\chi^{(3)}$ by $\chi^{(3)} = 4\epsilon_0 c n_2 n_0^2/3$, where ϵ_0 denotes the permittivity of vacuum. The coefficient n_2 is usually positive, leading to an increase of the refractive index in the presence of intense radiation. We discuss first the monochromatic (or continuous) regime for which the intensity of the laser does not depend on time.

The intensity of a beam being usually highest on axis, at the center of the beam, an effect of curvature of the wavefront similar to that of a lens is involved, with the difference that here the effect is cumulative and can lead, in the absence of other saturating effects, to a catastrophic collapse of the beam on itself. This effect is represented in a diagrammatic way in Fig. 3(a).

The characteristic length L_{SF} for self-focusing is defined as the length over which the cumulated nonlinear phase, measured by the B -integral ($B \equiv k_0 \int_0^z n_2 I \, dz$), varies by a factor of one. It is expressed as a function of the peak intensity I_0 :

$$L_{\text{SF}} = \frac{1}{n_2 k_0 I_0}. \quad (3)$$

It should be noted that the significant parameter for self-focusing is the initial power P_{in} of the beam and not the initial intensity (power per unit of area). Self-focusing overcomes diffraction and leads to collapse only if the input peak power P_{in} exceeds a critical threshold (Marburger, 1975):

$$P_{\text{cr}} \equiv 3.72 \lambda_0^2 / 8\pi n_0 n_2. \quad (4)$$

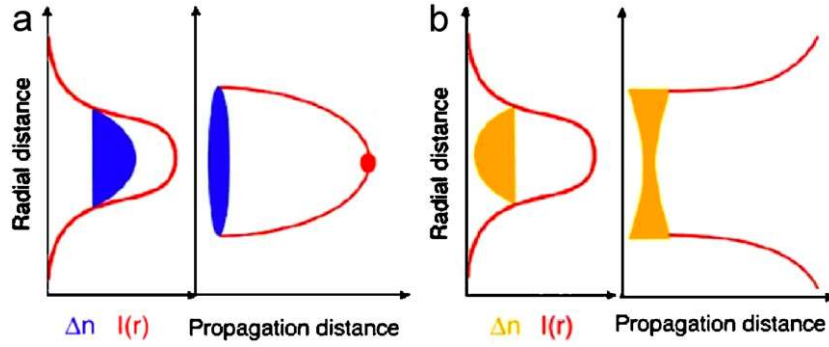


Fig. 3. (a) Self-focusing of a beam by optical Kerr effect. The refractive index of the medium depends on the intensity of the laser and acts as a lens by making convergent an initially collimated beam. Self-focusing prevails over diffraction when the power of the beam exceeds a critical power P_{cr} ($P_{cr} = 3.2$ GW for air at the wavelength of 800 nm) and leads, in the absence of other nonlinear effects, to the collapse of the beam on itself. (b) Defocusing of the beam by the presence of a plasma. The ionization of the medium initially takes place in the center of the beam, where the intensity is most significant. The creation of an under-dense plasma decreases the local index of the medium, which causes beam defocusing.

Eq. (4) is valid for a specific beam shape, sometimes called the Townes beam (Chiao et al., 1964), for which diffraction and self-focusing are balanced. This critical power will be used as a reference throughout this paper; beams with another shape undergo collapse when their power exceeds a threshold usually larger than that given by Eq. (4) (Fibich and Gaeta, 2000). For example the coefficient 3.72 must be replaced by 3.77 to obtain the critical power corresponding to a Gaussian beam.

The propagation length of the self-focusing beam until collapse L_c is rather well approximated by a semi-empirical formula (Marburger, 1975; Dawes and Marburger, 1969):

$$L_c = \frac{0.367 L_{DF}}{\sqrt{[(P_{in}/P_{cr})^{1/2} - 0.852]^2 - 0.0219}}, \quad (5)$$

where L_{DF} is the Rayleigh length of the beam. It is thus seen that L_c , like L_{DF} , varies like the square of the diameter of the laser beam. Eq. (5) is valid for the collapse of Gaussian beams with moderate input powers in purely Kerr media; see e.g., Fibich (1996) for estimations of the collapse position of non-Gaussian beams.

For a collimated beam, at a pulse wavelength of 800 nm, having a power $P_{in} = 10P_{cr}$ and a waist of $w_0 = 0.5$ cm, the distance to collapse L_c is approximately 15.6 m. At higher input powers ($100 P_{cr}$), the validity of Eq. (5) breaks down due to modulational instabilities. A transition from a $1/\sqrt{P_{in}}$ to a $1/P_{in}$ scaling was observed both numerically and experimentally (Fibich et al., 2005a).

In the case of a convergent beam (divergent), the position of the collapse $L_{c,f}$ moves before the position of the linear focus of the lens according to

$$\frac{1}{L_{c,f}} = \frac{1}{L_c} + \frac{1}{f}, \quad (6)$$

where f denotes the focal length of the lens.

The critical power given by Eq. (4) is also used as a reference value for ultrashort pulses. In this case, other physical effects as, e.g., group velocity dispersion, or multiphoton absorption and plasma defocusing which will be introduced later, may generally come into play during the self-focusing process and make the scenario of collapse more complex. An additional dimension, here time, is generally involved with these effects, adding a freedom degree to the system. Depending on which physical effect comes into play, the question of the occurrence of collapse in the sense of a singularity at finite distance is still an open question (see Fibich and Papanicolaou, 1999 for a review). For ultrashort pulses, GVD delays the onset of self-focusing since it redistributes the pulse along the temporal axis (Luther et al., 1994a). Due to the absence of any general expression for the critical power for self-focusing in the nonmonochromatic case, it is the habit to consider the peak power of the pulse P_{in} (its energy divided by the duration of the pulse) and to compare this value with that given by Eq. (4) in the monochromatic case. For 100 fs infrared laser pulses, this approach indicates that while the condition $L_{GVD} \gg L_c$ is fulfilled, self-focusing dominates diffraction if $P_{in} > P_{cr}$.

In air at atmospheric pressure, for a laser in the near infra-red domain ($\lambda_0 = 800$ nm), $P_{\text{cr}} = 3.2$ GW, which corresponds to a pulse energy of about 0.4 mJ, for a 100 fs laser pulse. It is thus clearly seen that femtosecond pulses launched in air easily reach a strongly nonlinear propagation regime: if the peak power P_{in} is smaller than P_{cr} , the beam diffracts and the pulse duration increases upon propagation, so that the intensity peak quickly becomes negligible. On the other hand, for $P_{\text{in}} > P_{\text{cr}}$, self-focusing occurs, the pulse duration decreases instead of increasing, the intensity remains high and the propagation regime is strongly nonlinear.

In a real medium, the collapse can be arrested by various mechanisms discussed below. While the relevant mechanism for the arrest of collapse before filamentation in air is multiphoton absorption and the associated plasma defocusing, the physical effect responsible for the arrest of collapse in condensed media is still a debated topics. Among the possible candidates, there are group velocity dispersion (Zharova et al., 1986; Chernev and Petrov, 1992; Rothenberg, 1992b; Fibich et al., 1995; Ranka et al., 1996; Fibich et al., 2003), saturation of n_2 (Fibich and Papanicolaou, 1999; Aközbek et al., 2000; Couairon, 2003a), nonlinear losses (LeMesurier, 2000; Polyakov et al., 2001; Zharova et al., 2002). Vectorial or nonparaxial effects can also, in principle, arrest collapse (Fibich and Papanicolaou, 1997; Fibich and Ilan, 2001a, b).

1.2.5. Self-phase modulation

The temporal variation of the laser intensity involves a temporal variation of the refraction index as $n = n_0 + n_2 I(r, t)$, which in turn corresponds to the appearance of new frequencies in the spectrum of the laser pulse. This effect is called self-phase modulation (SPM). The simplest model accounting for the evolution of the temporal phase links the instantaneous frequencies to the pulse intensity as:

$$\omega(t) = -\frac{\partial \phi}{\partial t} \sim \omega_0 - \frac{n_2 \omega_0}{c} z \frac{\partial I(r, t)}{\partial t}. \quad (7)$$

The generation of new frequencies thus depends on the slope of the pulse $\partial I / \partial t$, the propagation distance z in the Kerr medium and its nonlinear index coefficient n_2 . In a purely Kerr medium, the front part of the pulse generates redder frequencies, the back part bluer frequencies. Self-phase modulation plays a significant role in the appearance of a continuum of emission covering the visible and extending towards the infra-red domain.

1.2.6. Raman contribution to the Kerr effect

The optical Kerr effect can originate from two contributions, a pure electronic response with a time constant below 1 fs, and a slower response due to the nuclear motion (Stolen et al., 1989; Stolen and Tomlison, 1992). The optical Kerr coefficient is therefore dependent on the pulse duration. If the pulse duration is shorter than the Raman response time, only the electronic (quasi-instantaneous) response of electrons contributes. This is the case in air for a 10 fs pulse. In gaseous or liquid media with anisotropic molecules, orientation of the molecules in response to the incident fields makes a major contribution to the nonlinear index. Typical response times due to reorientation of the molecules are of the order of picoseconds to several hundred of picoseconds or even more in liquids.

1.2.7. Self-steepening

Laser pulse self-steepening occurs because the velocity of the peak intensity of the pulse is smaller than that of the trailing edge of the pulse, due to the intensity dependent change of refractive index $\Delta n = n_2 I$ (DeMartini et al., 1967). For Δn positive and instantaneous, the light velocity decreases with increasing laser intensity. Starting from a Gaussian pulse, the peak is slowed down with respect to the pulse group velocity while the trailing part catches up with the peak, leading to the formation of a steep edge in the trailing part of the pulse. This effect leads to a faster self-focusing of the trailing part of the pulse (in comparison with the leading part) and to an increase of the bluer frequencies generated in the steep trailing part (see e.g. Rothenberg, 1992b, and Fig. 4). Self-steepening also leads to asymmetric pulse splitting (Fibich and Papanicolaou, 1997; Ranka and Gaeta, 1998).

1.2.8. Photo-ionization

Filamentation requires a nearly lossless initial Kerr region followed by an abrupt onset of multiphoton absorption and plasma defocusing. For filamentation to occur, it is therefore necessary that the energy of the laser photons undergoing filamentation is a small fraction of the ionization potential U_i of the gas. For a small ionization potential $U_i < 3\hbar\omega_0$, the gradual attenuation of the pulse (due to two-three photon absorption) is too important to allow the formation of a narrow filament. The condition $\hbar\omega_0 \ll U_i$ is fulfilled for instance in air at 800 nm: $\hbar\omega_0 = 1.5$ eV, $U_i \sim 12$ eV since

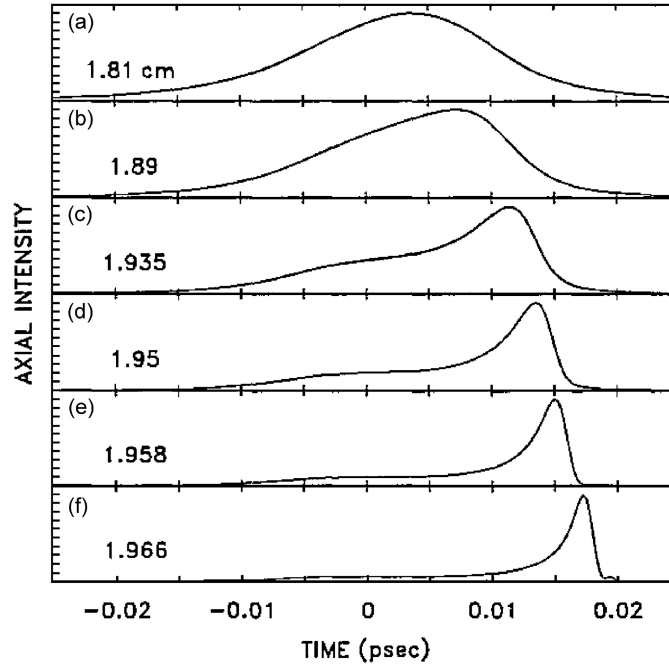


Fig. 4. Calculated intensity on axis (normalized units) as a function of time for a self-focusing and self-steepening pulse in a normally dispersive medium (xenon gas at 40 atm). The propagation distances are indicated on the graphs in cm. From Rothenberg (1992b).

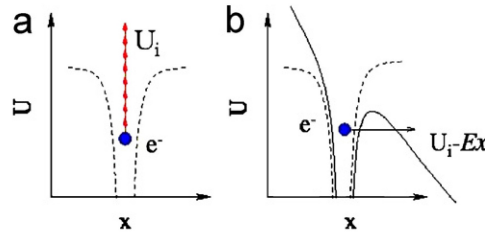


Fig. 5. Schematic diagram of: (a) multiphoton and (b) tunnel ionization.

oxygen is the first ionized constituent. Ionization of the medium requires the simultaneous absorption of a large number of photons $\sim U_i/\hbar\omega_0$, a process highly improbable at low intensity. However, when the beam approaches collapse, the pulse intensity increases enormously and ionization becomes possible. Ionization steps in very abruptly because its probability strongly depends on the light intensity. Photo-ionization or optical-field-ionization (OFI) covers both the so-called multiphoton and tunnel regimes: multiphoton ionization (MPI) is described by perturbation theory and is the first process to set in, in terms of increasing intensities. The ionization rate scales as I^K where $K \sim 8$ for air (oxygen) at 800 nm. Tunnel ionization occurs at higher intensities. It is described by the tunnel escape of an electron across the barrier formed by the combined action of the electromagnetic field and the Coulomb potential tying the electron to the nucleus (see Fig. 5). At even higher intensities, the potential barrier is suppressed and the electron is born in the ionization continuum at each oscillating cycle of the field. The corresponding attenuation of the beam is mentioned below in Section 1.2.11.

1.2.9. Plasma defocusing

The generation of a plasma involves a local reduction in the refraction index (Feit and Fleck, 1974), according to the law

$$n \simeq n_0 - \frac{\rho(r, t)}{2\rho_c}, \quad (8)$$

where $\rho(r, t)$ is the density of free electrons and $\rho_c \equiv \epsilon_0 m_e \omega_0^2 / e^2$ denotes the value of the critical plasma density above which the plasma becomes opaque (m_e and e denote the electron mass and charge, $\rho_c \sim 1.7 \times 10^{21} \text{ cm}^{-3}$ at 800 nm). The characteristic length L_{PL} for plasma defocusing is defined as the length over which the B -integral ($B = -k_0 \int_0^z \rho / (2n_0 \rho_c) dz$) corresponding to the integrated nonlinear phase induced by a fully ionized plasma ($\rho = \rho_{\text{at}}$) only changes by a factor of one. Following the same reasoning as for the derivation of Eq. (3), L_{PL} is expressed as a function of the neutral atom density ρ_{at} :

$$L_{\text{PL}} = \frac{2n_0 \rho_c}{k_0 \rho_{\text{at}}}. \quad (9)$$

For a weakly ionized medium of density ρ , the length scale for plasma defocusing is therefore $L_{\text{PL}} \rho_{\text{at}} / \rho$. For a gas of density $\rho_{\text{at}} = 2 \times 10^{19} \text{ cm}^{-3}$, and an electron density $\rho = 10^{16} \text{ cm}^{-3}$, we obtain $L_{\text{PL}} \sim 22 \mu\text{m}$ and $L_{\text{PL}} \rho_{\text{at}} / \rho \sim 44 \text{ cm}$. The reduction in the refraction index acts as a divergent lens, preventing the complete collapse of the beam on itself. This effect is schematically shown in Fig. 3(b). However, it should be stressed that the forward part of the pulse generates the electron plasma which in turn defocuses mainly the trailing part of the pulse. In addition to the purely spatial aspect illustrated in Fig. 3, plasma defocusing acts locally in time and leads to a strong asymmetry in the pulse temporal profile, which is partly at the origin of the dynamics in filamentation and temporal compression effects discussed in Sections 1.3.6, 3.2.4 and 4.1.5.

When an electron plasma is generated in the wake of the pulse, this also contributes to a spectral broadening and shift towards blue in the leading part of the pulse, similarly to the phenomenon of spectral broadening in a laser breakdown plasma (Yablonovitch, 1974a, b). In the presence of ionization, the simplest model relates the instantaneous frequency to the time dependent plasma density ρ as

$$\omega(t) = -\frac{d\phi}{dt} \sim \omega_0 + \frac{\omega_0 z}{c} \left(-n_2 \frac{\partial I(r, t)}{\partial t} + \frac{1}{2n_0 \rho_c} \frac{\partial \rho(r, t)}{\partial t} \right). \quad (10)$$

This is the counterpart of Eq. (7) when self-phase modulation induced by the plasma is also included. Plasma induced SPM was proposed as a mechanism to generate pulses of a few optical cycles (Yablonovitch, 1974a).

In addition to self-phase modulation, the plasma generated in the wake of the propagating pulse was also recently shown to modify the dispersive properties of the medium (Koprinkov, 2004). In particular, in air at infrared wavelength, the changes in group velocity dispersion induced by the plasma are sufficient to counteract locally the normal GVD and leave the medium nearly nondispersive.

1.2.10. Losses due to plasma absorption

Electrons produced by multiphoton or tunnel ionization can be further accelerated by the electric field of the remaining part of the pulse in an inverse Bremsstrahlung effect. If the electrons acquire enough kinetic energy, they give rise to a second generation of electrons by impact ionization of other molecules or atoms, in an avalanche like process. This process is at the origin of electric breakdown of long pulses in air or transparent solids, preventing long range propagation of laser energy.

To first approximation, plasma absorption can be formulated following the Drude model, in which the electron plasma is treated as a fluid. The collective electron velocity $\mathbf{v}$ is supposed to respond to the optical field and, consequently, the total current density $\mathbf{j} = -e n_e \mathbf{v}$ is governed by

$$\frac{d\mathbf{j}}{dt} = \frac{e^2}{m_e} \rho \mathbf{E} - \frac{\mathbf{j}}{\tau_c}, \quad (11)$$

where τ_c is the electron collision time. In the frequency domain, this leads to a current density:

$$\mathbf{j} = \frac{\tau_c (1 + i\omega\tau_c)}{1 + \omega^2\tau_c^2} \frac{e^2}{m_e} \rho \mathbf{E}. \quad (12)$$

Absorption of the laser pulse due to the plasma is given by

$$\frac{1}{2} \text{Re}(\mathbf{j} \cdot \mathbf{E}^*) = \frac{\tau_c}{1 + \omega^2\tau_c^2} \frac{e^2}{2m_e} \rho |\mathbf{E}|^2 \equiv \sigma \rho I, \quad (13)$$

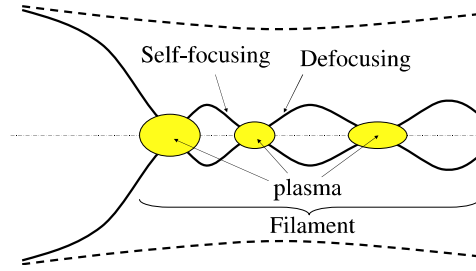


Fig. 6. Schematic representation of the focusing–defocusing cycles undergone by the intense core of the beam. The solid curves indicate the diameter of the intense core. The filamentation length is the distance covered by these cycles. The dashed line indicates the root mean square radius of the full beam.

where $I \equiv \frac{1}{2} \epsilon_0 c n_0 |\mathbf{E}|^2$ and σ denotes the cross section for inverse Bremsstrahlung (Raizer, 1965) which reads

$$\sigma = \frac{e^2}{\epsilon_0 m_e c n_0} \frac{\tau_c}{(1 + \omega^2 \tau_c^2)}. \quad (14)$$

1.2.11. Losses due to multiphoton absorption

Optical field ionization (OFI) is associated with energy losses. The beam attenuation due to multiphoton absorption can be generally described as a dissipative current J_i satisfying:

$$\mathbf{J}_i \cdot \mathbf{E} = \sum_k \rho_k W_k^{\text{OFI}}(I) U_{i,k}, \quad (15)$$

where ρ_k , $W_k^{\text{OFI}}(I)$ and $U_{i,k}$ denote the atom or ion density, the field dependent ionization rate and the ionization potential of species k in the medium. For multiphoton ionization of a single constituent, the ionization rate scales as $W^{\text{OFI}}(I) = \sigma_K I^K$, where $K \equiv \langle U_i / \hbar \omega_0 + 1 \rangle$ denotes the number of photons at the frequency ω_0 necessary to liberate an electron, and σ_K the ionization cross section. The typical length for multiphoton absorption, defined as the length over which a pulse of intensity I is attenuated by a factor $[(K + 1)/2]^{1/(K-1)}$, reads:

$$L_{\text{MPA}} = \frac{1}{2K \hbar \omega_0 \sigma_K I^{K-1} \rho_{\text{at}}}. \quad (16)$$

For a filamented pulse at 800 nm in air ($K = 8$, $\sigma_8 = 3.7 \times 10^{-96} \text{ cm}^{16} / \text{W}^8 / \text{s}^1$), typical values are $L_{\text{MPA}} \sim 160 \mu\text{m}$ for $I = 5 \times 10^{13} \text{ W/cm}^2$ and $L_{\text{MPA}} \sim 12.6 \text{ m}$ for $I = 10^{13} \text{ W/cm}^2$.

1.2.12. Refocusing cycles

It is primarily the combined action of the optical Kerr effect, multiphoton absorption and ionization which are at the origin of the formation of filaments preserving an average diameter of about $100 \mu\text{m}$ over long distances in air, even if other effects, like diffraction, group velocity dispersion, self-phase modulation and pulse self-steepening contribute to this self-guiding process. This process is shown in Fig. 6. A pulse with peak power above P_{cr} should undergo a collapse. However, when the beam becomes sufficiently intense, MPA attenuates the core of the beam and a plasma is generated near the collapse location, which defocuses the beam core. Once defocused, the beam may still contain a power above critical and undergo another focusing–defocusing cycle. The repetition of this scenario sustains a long range, self-guided propagation in the form of a filament. The interplay between self-focusing, MPA and ionization is therefore a highly dynamic process, with recurrent, aperiodic strings or spikes of ionization surging whenever the beam starts collapsing again (Mlejnek et al., 1998a). The recurrence of these cycles depends among other things on the peak power of the input pulse.

At this point, it should be stressed again that *self-guiding* or *self-channeling* is nothing else than a terminology meaning that no external waveguide sustains this long range propagation. By these terms, it is not assumed by any means that the filament is a soliton-like structure or that there is a delicate balance between the index variations attributed to each physical effect. Rather, a dynamic competition between self-focusing and defocusing effects (plasma defocusing among others) involves successive focusing–defocusing cycles in the global dynamics.

These physical effects couple space and time since self-phase modulation (the temporal counterpart of self-focusing), simultaneously competes with group velocity dispersion, self-steepening and shock formation due to plasma defocusing. This leads to strong pulse distortions during propagation. Multiphoton and plasma absorption also indirectly participate in this complex coupling by attenuating the peak intensity.

1.2.13. Modulational instability

When the laser pulse power is well above P_{cr} , the beam starts collapsing over many spots, each forming its own filament. The appearance of many filaments is related to the growth of components of the beam intensity profile with specific transverse spatial frequencies. The growth rate k_i of spatial modulation structures is a function of the transverse wave number of the perturbations which reads as $k_i(k_{\perp}) = (k_{\perp}/2k_0)\sqrt{8\pi I_0/P_{\text{cr}}^{\text{G}} - k_{\perp}^2}$, where $P_{\text{cr}}^{\text{G}} \equiv \lambda_0^2/2\pi n_0 n_2$ denotes an estimation of the critical power for self-focusing associated to a Gaussian beam and I_0 denotes the intensity of the laser field. The maximum growth rate of spatial modulation structures occurs at a transverse wavenumber $k_{\perp}^{\text{opt}} = 2\sqrt{\pi I_0/P_{\text{cr}}^{\text{G}}}$ (Campillo et al., 1973, 1974). After a growth of the instability waves as $\sim \exp[k_i(k_{\perp}^{\text{opt}})z]$, saturation occurs in the form of filaments with a transverse spacing between the filaments equal to $d_{\text{fil}} = 2\pi/k_{\perp}^{\text{opt}} = \sqrt{\pi P_{\text{cr}}^{\text{G}}/I_0}$. Assuming a regular filamentation pattern, one can roughly estimate the individual power that a filament should be able to carry as $P_{\text{fil}} = 2\pi \int_0^{d_{\text{fil}}} I_0 r dr \sim (\pi^2/4) P_{\text{cr}}^{\text{G}}$ (Couaïron and Bergé, 2000). From these energy considerations, a beam with power P_{in} well above critical leads to a number of filaments $N \sim P_{\text{in}}/P_{\text{fil}}$. It should not be forgotten that these considerations rely on several assumptions, among which the fact that the instability analysis is performed on plane and monochromatic waves. For Gaussian pulses, the threshold for modulational instability initiated by noise was found to lie around $100 P_{\text{cr}}$ (Fibich et al., 2005a). In addition, the nonlinear saturation of the growth of filaments after the linear stage of the instability is assumed not to destroy the filamentation pattern. Therefore, the estimated number of filaments should not be expected to be accurate. The above considerations are not qualitatively modified when other physical effects such as multiphoton ionization or absorption are accounted for; in this case the spatial growth rate is reduced with respect that obtained in a purely Kerr medium (Couaïron and Bergé, 2000; Marklund and Shukla, 2006).

1.2.14. Filaments in condensed media

Filamentation in condensed media is not only interesting for applications (see Section 5.2.8), but also constitute a remarkable system where all the physical effects and properties presented above are observed on a reduced scale. Dense transparent dielectrics have a nonlinear refraction index about 3 orders of magnitude larger than in gases. As a result, the critical power for self-focusing is in the Megawatt range while it is in the Gigawatt range for gases. Only $1 \mu\text{J}$ energy is required for a laser pulse of 100 fs in order to self-focus the beam in a solid. Remarkably, the peak intensity in the filament is almost the same as in gases. As will be shown later, a filament carries about one critical power for self-focusing. From the peak intensity and the carried power, the filament diameter is estimated to be about a few microns to a few tens of microns, i.e. 30 times narrower than in gases (Yablonovitch and Bloembergen, 1972; Tzortzakis et al., 2001d). It is worth underlining here that the ionization mechanism generating free electrons in gases is replaced in solids by the promotion of electrons from valence to conduction band. The electron density reaches typically 10^{18} – 10^{20} cm^{-3} which is three orders of magnitude larger than that reached in air but still corresponds to a low density plasma.

The Rayleigh length of an 800 nm laser beam corresponding to $w_0 = 10 \mu\text{m}$ is about $390 \mu\text{m}$. With the same intensity but a much higher electron density, energy losses are much higher in condensed media. This quickly exhausts the laser energy reservoir and filamentation stops because the peak power of the pulse quickly goes below the critical power for self-focusing.

Another important difference between condensed media and gases is the higher GVD. As will be shown in Section 1.2.15, GVD is an efficient mechanism to arrest the collapse.

1.2.15. Arrest of collapse

From the previous presentation, one might draw the conclusion that plasma defocusing alone is responsible for the arrest of collapse that would occur in a purely Kerr medium when the input beam power is above the critical power for self-focusing. Several mechanism, however, are able to play this role. Among them, there are saturation of the nonlinear index by a higher order (defocusing) nonlinearity (Aközbek et al., 2000; Couaïron, 2003a), nonlocal effects

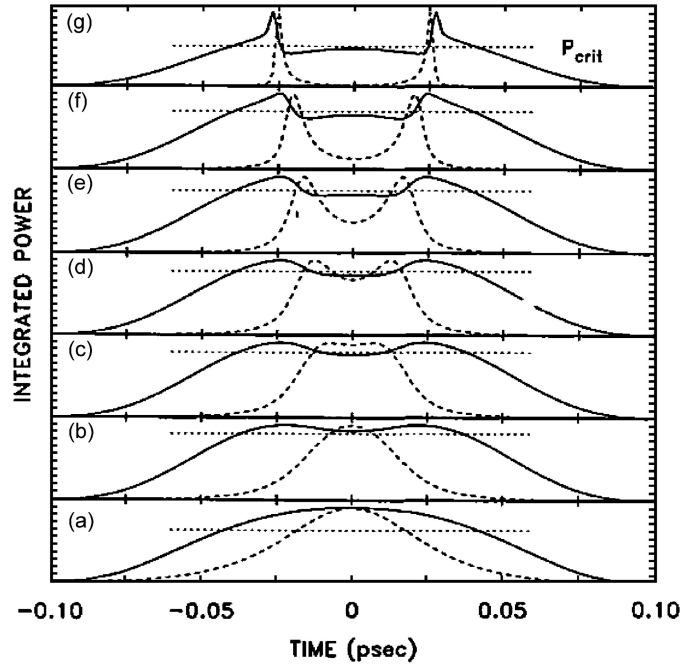


Fig. 7. Pulse splitting due to the competition between the optical Kerr effect and GVD in a 40 atm Xe gas. The 600 nm input pulse has a FWHM duration of 50 fs and peak power of 45 MW. The beam of 500 μm FWHM is focused with $f = 25$ cm. The solid curves show the spatially integrated power at successive propagation distances. (a) 17.53 cm; (b) 18.6 cm; (c) 18.79 cm; (d) 18.86 cm; (e) 18.92 cm; (f) 18.98 cm; (g) 19.04 cm. The dashed curves are the axial intensities. The dotted lines indicate the critical power level (15 MW). According to [Rothenberg \(1992a\)](#).

(plasma defocusing, Raman Kerr effect) ([Bang et al., 2002](#); [Turitsyn, 1985](#)) or nonlinear absorption ([Vlasov et al., 1989](#); [Kosmatov et al., 1991](#); [Dyachenko et al., 1992](#); [LeMesurier, 2000](#); [Couairon et al., 2006c](#)).

While in gases, GVD is too small to play a significant role in the arrest of collapse, several authors have pointed out that GVD, which is much stronger in solids, leads to pulse splitting into two components moving in opposite directions in the frame of the input pulse (see [Fig. 7](#)) ([Zharova et al., 1986, 2002](#); [Chernev and Petrov, 1992](#); [Rothenberg, 1992a](#); [Fibich et al., 1995, 2003](#); [Fibich and Papanicolaou, 1997](#)). With GVD alone, these authors describe pulse propagation beyond the nonlinear focus. [Ranka et al. \(1996\)](#) have shown that GVD is an efficient mechanism arresting the collapse. This mechanism, however, is relevant for pulses having a sufficiently low peak power, since the critical power threshold for collapse in dispersive media, determined analytically by [Luther et al. \(1994a\)](#) increases with dispersion. We remind that for pulses with powers largely exceeding the critical power for self-focusing, multiphoton absorption is an even more efficient mechanism than GVD in arresting the collapse and it leads to multiple splitting beyond the nonlinear focus (see [Vlasov et al., 1989](#); [Kosmatov et al., 1991](#); [Dyachenko et al., 1992](#); [LeMesurier, 2000](#); [Couairon et al., 2006c](#)). Pulse splitting is therefore not systematically induced by GVD, although GVD as well as space–time focusing and self-steepening must often be taken into account so as to correctly describe the propagation beyond the nonlinear focus in the slowly varying envelope approximation ([Ranka and Gaeta, 1998](#)).

1.3. Properties of light filaments

1.3.1. Robustness of filaments

A remarkable feature of filaments is their universality: they appear in many transparent media whatever their solid, liquid or gaseous nature, provided that the photon energy is much less than the fundamental electronic transition energy. This suggests that a robust attractor of the propagation dynamics might exist in the nonlinear regime. Filaments are even regenerated when a central stopper blocks their propagation ([Courvoisier et al., 2003](#); [Kolesik and Moloney, 2004b](#); [Dubietis et al., 2004b](#); [Liu et al., 2005e](#)). This property was called self-healing by [Kolesik and Moloney \(2004b\)](#). Filaments are actually rebuilt from the laser energy reservoir ([Mlejnek et al., 1999a](#)). As a result filaments can traverse

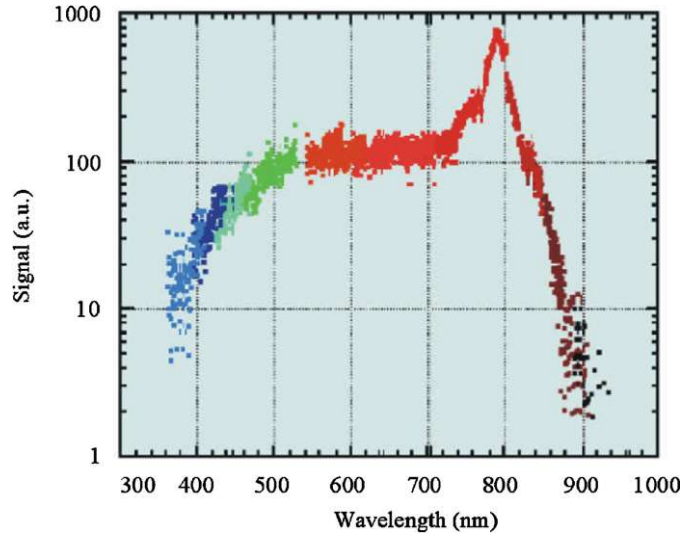


Fig. 8. Spectrum of a femtosecond laser beam ($\lambda_0 = 800$ nm; $\tau_{\text{FWHM}} = 70$ fs, $P_{\text{in}} = 3$ TW) after propagation over a distance of 10 m in air. The fast fall between 800 and 900 nm corresponds to the fall of the detectivity of the measuring apparatus (see Kasparian et al., 2000b).

adverse atmospheric conditions such as foggy or rainy atmosphere (Méchain et al., 2005b). This property is interesting for all applications where a large amount of laser energy must be conveyed over extended distances in the sky (see Section 5).

1.3.2. Long range propagation

One of the most spectacular properties of filamentation is its ability to convey high intensities over extended distances. Peak intensities exceeding 10^{12} W/cm² have been observed over several kilometers via filamentation (Méchain et al., 2005a). Energy losses during filamentation in air are minimized because: (a) the intensity is always maintained at the verge or below the value corresponding to ionization threshold via the clamping mechanism; (b) the short pulse duration prevents significant pulse attenuation through its interaction with the generated plasma (via inverse Bremsstrahlung).

1.3.3. Spectral broadening

Filamentation is accompanied by a strong spectral broadening, which depends on the strong reorganization of the temporal shape of the pulse. Several effects combine to cause spectral broadening: self-phase-modulation, self-steepening and ionization of the medium. In particular, the sharp trailing part of the pulse is due to self-steepening as well as ionization, which add preferentially new frequencies on the high energy side of the spectrum (see Fig. 8). Thus, an initial pulse centered in the near infrared at 800 nm partially converts into a continuum covering the entire visible part of the spectrum when propagating in air. On the low energy side, continuum generation has been observed up to 5 μ m in the IR (Kasparian et al., 2000b) but with a much lower intensity. The conversion rate into white light varies from a few tenth of percent for an initially collimated beam up to an appreciable fraction of the initial energy, for a convergent beam. A similar phenomenon forming a so-called white light laser was reported in transparent solids (Petit et al., 2000b).

1.3.4. Intensity clamping

As we have seen, filamentation results predominantly from a dynamic competition between the optical Kerr effect, multiphoton ionization and associated multiphoton absorption. Multiphoton ionization sets in once the beam intensity reaches a sufficiently high value. The multiphoton ionization rate scales first as I^K , with K , the number of photons simultaneously absorbed, typically greater than 5 for infrared wavelengths. The generated electron density has therefore a threshold like response, which saturates self-focusing locally and limits the peak intensity inside the filament by defocusing the beam. This simple idea allows an estimation of the clamping intensity according to Braun et al. (1995),

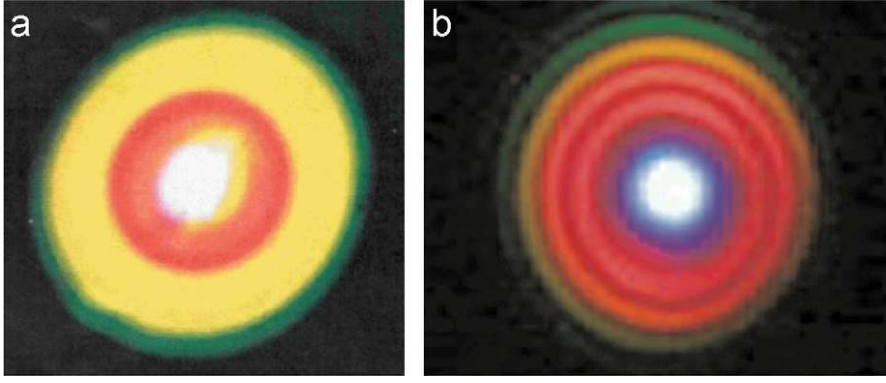


Fig. 9. Conical emission accompanying a self-guided pulse. (a) A central white core (the filament) is surrounded by Newton's rings having a divergence of the order of the mrad (Nibbering et al., 1996). (b) Conical emission measured at 25 m from a 5 mJ, 45 fs, 800 nm pulse (Chin, 2004).

Kasparian et al. (2000a), Becker et al. (2001a):

$$n_2 I = \frac{\rho(I)}{2\rho_c}. \quad (17)$$

In air, by using the rough estimate $\rho(I) \sim \sigma_K I^K \rho_{\text{at}} t_p$, where ρ_{at} denotes the density of neutral atoms, we obtain

$$I \sim \left(\frac{2n_2 \rho_c}{\sigma_K t_p \rho_{\text{at}}} \right)^{K-1}, \quad (18)$$

which yields the clamping intensity $I \sim 1.8 \times 10^{13} \text{ W/cm}^2$ for a pulse duration $t_p = 100 \text{ fs}$ at 800 nm. The filament contains a fixed amount of power, roughly equal to P_{cr} (Braun et al., 1995). Following Yablonovitch and Bloembergen (1972), the transverse size of the filament corresponding to the above power ($P_{\text{cr}} = 3.2 \text{ GW}$) and intensity given by Eq. (18) is $w_0 = 105 \mu\text{m}$.

Liu et al. (2002c) have found also evidence of the clamping phenomenon from an analysis of the supercontinuum generated by filamentation of 170 fs, 800 nm laser pulses in condensed media (water, chloroform and glass). The maximum blueshift due to self-phase modulation was found to be independent of the pulse energy above a critical value, which was interpreted from Eq. (10) as a manifestation of the clamping phenomenon.

1.3.5. Conical emission of white light and colored rings

The white light beam generated during filamentation in gases or condensed media generally consists of a white central part surrounded by a rainbow-like conical emission (see Fig. 9). The radial order of the spectral components is inverse of diffraction with bluer frequencies appearing on the outside rings. This phenomenon is a signature of filamentation. Several mechanism have been invoked to explain this still not entirely understood phenomenon: Cerenkov radiation (Golub, 1990; Nibbering et al., 1996), self-phase modulation (Kosareva et al., 1997a, b), four-wave mixing (Xing et al., 1993; Luther et al., 1994b), X-waves (Di Trapani et al., 2003; Conti et al., 2003; Faccio et al., 2006b). The reader is referred to Aumiler et al. (2005) for a review of tentative interpretations of conical emission in dense atomic vapors. Conical emission accompanying filamentation is discussed in more details in Section 2.1.7.

1.3.6. Pulse self-compression

In the filamentation process, several effects compete to reshape the pulse time profile. This leads to a significant shortening of the pulse duration. Recent calculations have shown that nearly single optical cycle pulses can be obtained in this way, starting with pulses of a few tens of femtoseconds duration. Fig. 10 shows a sequence of space time profiles of a femtosecond laser pulse at various stages as it progresses through a gas volume (Couairon et al., 2006a). Intense single cycle pulses in the visible or infrared are extremely important for producing attosecond pulses in the far UV via high-order harmonics generation (Brabec and Krausz, 2000). Pulse self-compression via filamentation may

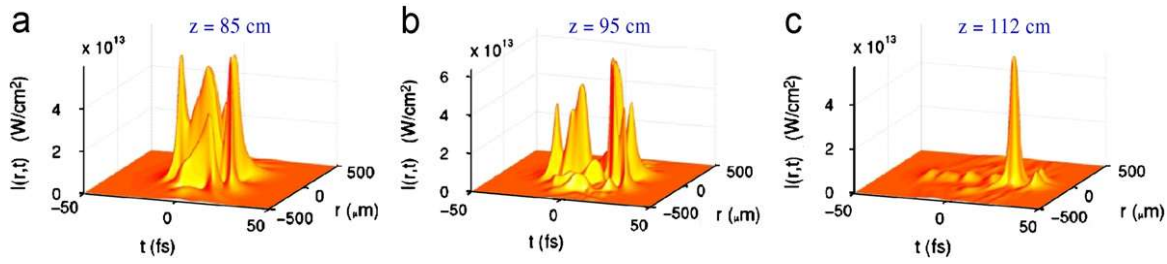


Fig. 10. Numerical simulations show the self-shortening and single cycle pulse generation of a 1 mJ, 25 fs infrared laser pulse undergoing filamentation in a gas cell filled with argon at 0.8 atm. According to Couairon et al. (2006a).

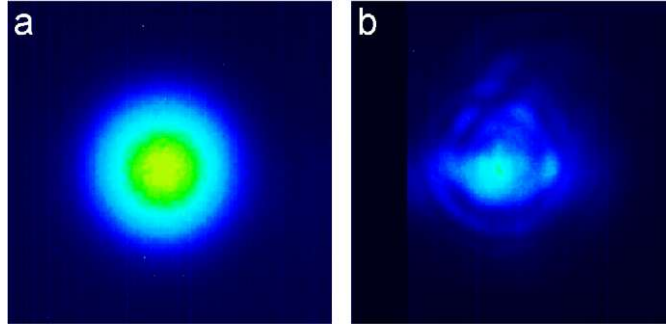


Fig. 11. Conical emission and laser pattern measured after 20 m of propagation in air from a 400 nm laser pulse. (a) Surrounding conical emission only, with the power spectrum close to that of the incident laser pulse removed by a colour filter. The external diameter of the blue ring is about 1 cm. (b) Strongly attenuated total laser beam at the same distance. From Prade et al. (2006a).

considerably simplify the experimental procedure to generate attosecond XUV pulses via high-harmonic generation with phase-stabilized single cycle laser pulses (Hauri et al., 2004; Chakraborty et al., 2006).

1.3.7. Pulse mode self-cleaning

By examining the far-field pattern of the conical emission surrounding a filament core generated from a 400 nm laser pulse, Prade et al. (2006a) noticed a significant improvement of the beam quality. The spatial conical emission associated with the filament corresponds to a single transverse radiation mode while the remnant of the incident pulse around 400 nm, which does not couple into the filament, exhibits a poorer beam quality (Prade et al., 2006a, b). This is shown in Fig. 11 which compares the spatial profile of the conical emission (with the wavelengths around the central laser frequency removed by a color filter placed at $z = 15$ m) with the total beam profile at the same distance comprising the remnant of the nonfilamentary part of the beam, the filament core and the much weaker conical emission. A striking feature is the excellent quality of the conical emission which takes the form of a single transverse mode. Filaments bear therefore remarkable properties in the time domain (pulse self-compression), frequency domain (spectrum broadening) and spatial domain (mode improvement) making it a unique source for applications (Chin et al., 2007).

1.3.8. Filaments in optical parametric amplifiers

A nice application of the unique properties of filaments is illustrated by the possibility to design optical parametric amplifiers where a filament acts as a pump source. Théberge et al. (2006a) have shown that a signal beam tunable over the entire visible spectrum is obtained with excellent beam quality. It corresponds to a diffraction limited Gaussian beam ($M^2 < 1.01$), translating the self-cleaning effect of a filament. When compared to an OPA system operating with the same laser as a pump except that it is not in a filament form, the signal beam has also a shorter pulse duration translating the self-compression effect of filaments. Finally, the signal beam exhibits a distinct improvement in beam stability interpreted as an effect of the intensity clamping phenomenon (see Fig. 12).

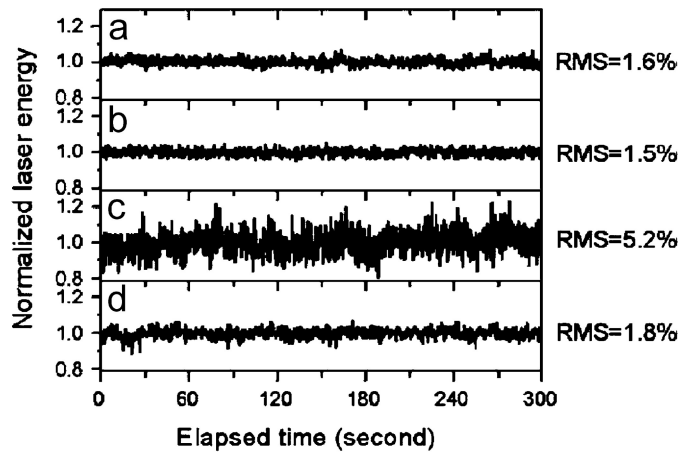


Fig. 12. Temporal evolution of energy per pulse normalized over the mean energy for (a) the infrared seed ($1.1 \mu\text{m} < \lambda < 1.6 \mu\text{m}$), (b) the near infrared pump pulse (807 nm), (c) the generated signal pulse for near infrared pump power (10 times below P_{cr} in air), and (d) the generated signal pulse for pump power of $2.5 P_{\text{cr}}$. The rms energy fluctuations are indicated on the right-hand side. From Th  berge et al. (2006a).

1.3.9. Superluminal velocity

One of the interesting properties of filamentation is that the pulse envelope undergoes strong temporal and spatial deformations. The motion of the peak intensity can even exceed the light velocity in the medium, a phenomenon called superluminal propagation.

Experimental evidence for superluminal propagation during filamentation has been reported by Tzortzakis et al. (2003) in an experiment on filament concatenation. It is possible to connect two spatially separated plasma strings by sending a sequence of two laser pulses with different curvatures and an appropriate delay. It was found that the chronological coincidence of the two pulses, which leads to the connection, could only be obtained by delaying the pulse with the smaller beam curvature (see Section 4.1.22 for details). This pulse which has undergone filamentation has an advance over its nonfilamented twin. This advance is exacerbated by introducing a thin fused silica plate in its path (see Fig. 13).

1.3.10. Generation of THz radiation

Plasma channels formed by femtosecond laser filaments in air radiate electromagnetic THz pulses (EMP), in a frequency range at the border region between optical sources and radio sources. This is an exciting property since it gives rise to a coherent emission from an ubiquitous medium in a challenging part of the electromagnetic spectrum. Since the location of filaments can be varied to a large extent, it provides a coherent THz source at a remote, selected location. THz and sub-THz emission have important applications in medical and other domains.

Heterodyne detection of sub-THz radiation emitted radially by plasma strings after filamentation in air was recently reported (Tzortzakis et al., 2002). Very recently, Houard et al. (2007a) and D'Amico et al. (2007b) reported a stronger sub-THz radiation emitted by the filament in a narrow forward cone.

1.3.11. Two-colored filamentation and third harmonic generation

As we have seen, the nonlinear response initiating filamentation is the optical Kerr effect described by the real part of the third order susceptibility of the medium. Third harmonic generation, a process associated with the imaginary part of $\chi^{(3)}$ will also take place. As shown by Ak  zbek et al. (2002b) the generated coherent wave at $3\omega_0$ is coupled with the fundamental wave ω_0 via cross-phase modulation (the refractive index $n_2(3\omega_0)$ is modified by the field at ω_0 and vice versa). This leads to the formation of a UV filament at $3\omega_0$. Cross-phase modulation couples the fundamental and the third harmonic pulse together, imposing a constant phase difference of π inside the two-colored filament. The cofilamentation of the high intensity fundamental and third harmonic pulses over long distances opens the way to promising applications for atmospheric remote sensing (Th  berge et al., 2005a–d).

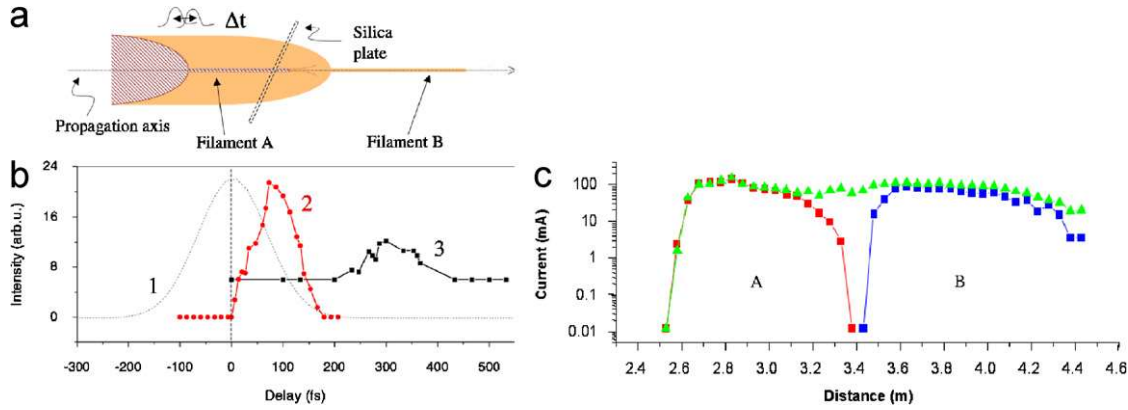


Fig. 13. (a) Schematic representation of the experiment. Two filaments A and B are generated along the same axis by using two pulses A and B with different beam convergence. A silica plate can be introduced in a region where only pulse A is a filament. For a well suited delay between pulses A and B, it is possible to connect the filaments in order to give rise to only one filament of constant density. (b) Curve 1 represents the second harmonic autocorrelation trace of the initial laser pulses. Curve 2 corresponds to the conduction signal measured at the junction region between the two filaments as a function of delay between pulses A and B: in the absence of the silica plate, it is necessary to advance pulse B by 100 fs (curve 2) in order to have chronological coincidence between pulses A and B in the junction region. Curve 3 is the signal measured in the presence of the silica plate. In this case, pulse B must be advanced by 300 fs. According to Tzortzakis et al. (2003). (c) Measurements of the conductivity of air for the two individual filaments A and B and the concatenated filament (Tzortzakis et al., 2003; Couairon et al., 2003).

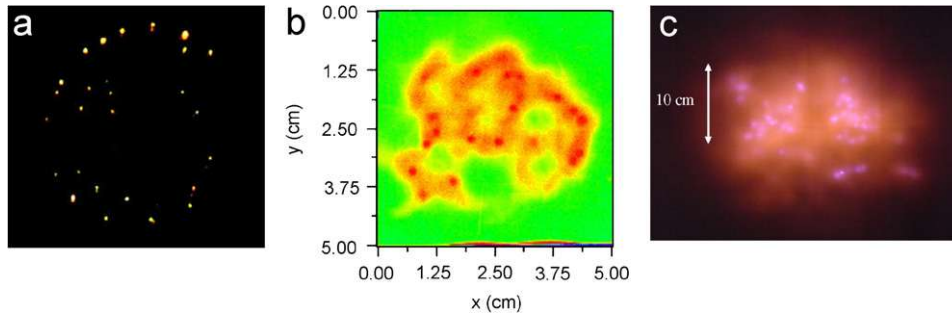


Fig. 14. (a) Image of a beam profile recorded on a screen at 50 m with the camera shutter opened immediately after the laser pulse. The spots corresponds to the long lasting phosphorescence of the screen induced by the filament spots. (b) Image of another beam profile recorded on the same screen at 50 m with the camera shutter open during the laser pulse. (c) Image of the beam profile recorded on a white screen at 600 m showing multifilamentary structure. For the three cases, the pulse of initial duration 100 fs has been stretched to 1.7 ps with a negative chirp (Méchain et al., 2005a).

1.3.12. Multiple filamentation

When the input peak power is far above the critical power for self-focusing, the modulational instability breaks up the beam into a large number N ($N \simeq P_{\text{in}}/P_{\text{cr}}$) of filaments (see Fig. 14(a)) (Vidal and Johnston, 1996). For large powers above $\sim 100P_{\text{cr}}$, multiple filamentation occurs at a distance scaling as $1/P_{\text{in}}$. For smaller powers, modulational instability can also lead to multiple filamentation but this occurs beyond the beam collapse at a distance scaling as $1/\sqrt{P_{\text{in}}}$ (Fibich et al., 2005a). These filaments, which grow from the small fluctuations in the beam intensity profile, can be seemingly randomly distributed in a given transverse plane z or organized into specific patterns (Mlejnek et al., 1999a; Méchain et al., 2004b; Dubietis et al., 2004c; Fibich et al., 2004; Schroeder et al., 2004). Each filament is likely to take a power about P_{cr} . When the power of an individual filament becomes smaller than P_{cr} , the filament restores the main part of its energy to a common photon bath; this energy reservoir can feed narrow filaments, which further leads to a new distribution of filaments becoming quickly random. The number of filaments N decreases gradually with the propagation distance due to energy losses by plasma or multiphoton absorption, until containing only one filament. Numerical simulations of the propagation must clearly be handled in $(3 + 1)$ dimensions in this case (three spatial dimensions + time) to capture the full dynamics of filamentation.

1.3.13. Energy reservoir

Fig. 14 shows two photographs of a screen intercepting the laser beam at a distance of 50 m. The laser beam contains about 200 mJ. After 50 m, more than 90% of the initial laser energy is still present inside the 1 cm diameter beam. Fig. 14(a) is recorded several ms after the pulse. The small spots still observable correspond to the long lasting fluorescence of the screen induced by the very intense filaments which have been formed. By contrast, Fig. 14(b) is recorded during the pulse. One sees again the bright spots corresponding to multiple filaments but in addition, one observes a more diffused illuminated area. This diffuse area forms an energy reservoir for the filament. Since the energy in each filament is at most one mJ, the total energy content of the filaments is around 30 mJ, a factor 10 below the total energy measured at that distance. Most of the energy is distributed in the reservoir. As can be seen, this energy tends to link different filaments through ducts joining adjacent filaments. The energy reservoir plays a very important role in the persistence of filament formation and regeneration over long distances as first pointed out by Mlejnek et al. (1999a).

2. Modeling filamentation

2.1. Model scenarii

The history of filamentation traces back to the early days of nonlinear optics (Askar'yan, 1962) where long narrow fluorescence or damage tracks were seen in solids and liquids (Hercher, 1964; Lallemand and Bloembergen, 1965; Shen and Shaham, 1965; Garmire et al., 1966). Two models were advocated to explain this phenomenon: one is the self-trapping model where a strict equilibrium between diffraction and self-focusing is proposed (Chiao et al., 1964, 1966; Talanov, 1964, 1965). This equilibrium is interpreted as a mode, often referred to as the Townes mode (Chiao et al., 1964), which was later shown to represent the self-similar blowup solution reached during the collapse of a self-focusing beam regardless of its initial shape (Landman et al., 1991, 1992; Fibich and Papanicolaou, 1999). The self-trapping model explains certain features such as the quality of the emerging mode and the associated considerable spectral broadening (Gustafson et al., 1969; Svelto, 1974) or the self-focusing distance (Kelley, 1965; Marburger and Dawes, 1968). The other model called the moving focus model considers the filament as an illusion, as explained in Section 2.1.1. A large part of the nanosecond pulse experiments were consistently explained by the moving focus model (Lugovoi and Prokhorov, 1968; Shen, 1984; Giuliano and Marburger, 1971; Loy and Shen, 1969, 1973; Korobkin et al., 1970). However, a more complete description (Shen, 1984) shows that a short pulse deforms into a horn-shape as it propagates, which can be viewed as a self-trapped appendix to the moving focus; this horn structure corresponds to the descending part of the pulse which sees a larger refractive index induced by the leading part. It is not astonishing that the same concepts have been invoked in the case of femtosecond laser pulses.

2.1.1. Moving focus model

In the moving focus model (Lugovoi and Prokhorov, 1968; Shen, 1984; Brodeur et al., 1997), the laser pulse is stacked into time slices that are considered to be independent from each other. This picture relies on the assumption that the influence of collapsed time slices on the subsequent propagation can be neglected. This approximation is valid when the physical effects coupling the various time slices together, such as e.g. GVD, remains weak. In air, the typical distance for GVD for a $t_p = 100$ fs infrared pulse is $z = 250$ m, much larger than the typical self-focusing and collapse distances. The approximation of independent time slices seems therefore justified at first sight. Each time slice contains a specific power. All the central slices of a pulse with peak power above critical are self-focused at distances that become larger as the corresponding power is closer to critical value. The slices with power below P_{cr} diffract. In this picture, the perception of a filament is constituted by the collection of nonlinear focii corresponding to the different temporal slices of the pulse comprised between the time slices t_A and t_B which correspond to a power P_{cr} on the ascending (resp. descending) slope of the pulse envelope. Fig. 15(A) shows the predicted locations of the moving focus as a function of time. Following the original model (see e.g. Lugovoi and Prokhorov, 1968; Shen, 1984), the velocities and propagation directions refer here to the laboratory frame and not to the reference frame moving at the pulse group velocity usually used in optics. Applying Eq. (5) to $P_{in} = P(\tau = t - z/v_g)$ yields a focus for each time slice in the pulse. Two branches appear: the upper branch starting from the earliest focus B describes the forward motion of the focus due to the forward part of the pulse $t_A < t \leq t_B$. This branch has a superluminal velocity. The lower branch BCD describes the backward motion of the focus from B to C, the time slice t_C with maximum power focusing at the closest position to the laser, followed by the forward motion from C to D, corresponding to the trailing part of the pulse.